

AT89C4051, timer NE555, LCD module (16x2 line) and a few discrete components. Timer NE555 is configured as a monostable multivibrator whose time period depends upon the combination of resistor R1 and capacitor C1. Trigger pin 2 of NE555 is pulled high via resistor R2. The LDR is connected along with resistor R2 to pin 2 of NE555 such that when the laser light falls on the LDR, pin 2 goes low to trigger NE555.

The output from pin 3 of NE555 is inverted by transistor T1 and fed to port pins P3.3 and P3.4 of the microcontroller. LED2 is connected to port pin P3.0 (pin 2) of the microcontroller. Data pins D0 through D7 of the LCD are connected to port pins P1.0 through P1.7 of the microcontroller, respectively. Control pins E, RS and R/W of the LCD are connected to port pins P3.2, P3.5 and P3.7, respectively. A 12MHz crystal connected between pins 4 and 5 of the microcontroller, along with two 22pF capacitors C4 and C5, generates the basic clock frequency. Power-on reset is derived with the combination of resistor R7 and capacitor C6. Switch S2 is used for manual reset.

An actual-size, single-side PCB for the RPM counter is shown in Fig. 3 and

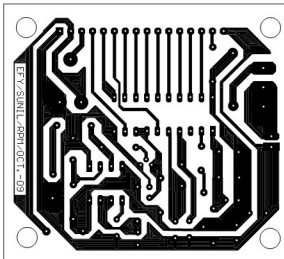


Fig. 3: Actual-size, single-side PCB for the RPM counter using microcontroller AT89C4051

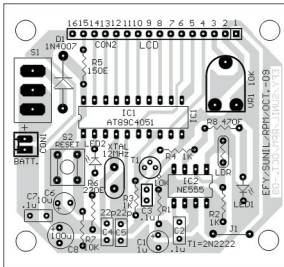


Fig. 4: Component layout for the PCB

PARTS LIST

Semiconductors:

IC1	- AT89C4051 microcontroller
IC2	- NE555 timer
T1	- 2N2222 npn transistor
D1	- 1N4007 rectifier diode
LED1, LED2	- 5mm LED
LCD	- 16x2 line

Resistors (all 1/4-watt, ±5% carbon):

R1-R4	- 1-kilo-ohm
R5	- 150-ohm
R6	- 220-ohm
R7	- 10-kilo-ohm
R8	- 470-ohm
VR1	- 10-kilo-ohm preset

Capacitors:

C1	- 1µF, 16V electrolytic
C2	- 0.01µF ceramic disk
C3, C7	- 0.1µF ceramic disk
C4, C5	- 22pF ceramic disk
C6	- 10µF, 16V electrolytic
C8	- 100µF, 16V electrolytic

Miscellaneous:

S1	- On/off switch
S2	- Push-to-on switch
X ₁	- 12MHz crystal
BATT.	- 6V battery
	- LDR

"Counting finished..." and goes off. The microcontroller stores the count and multiplies it by 60 to give the final RPM count.

5. The count is in hex format, so you have to convert it into decimal first. As the LCD accepts only ASCII values, the decimal values are converted into ASCII and shown on the LCD one by one.

6. Now if you press RST switch, the process repeats

Software

The software for the RPM counter is written in 'C' language and compiled using Keil µVision3 compiler. Burn the generated '.hex' file into the microcontroller using a suitable programmer. The source program is well commented and easy to understand.

The various functions are detailed below:

1. Main function initialises the timer, LCD, ports, etc, displays the message "RPM counter," clears the timer content and enables the external interrupt. It remains in the loop till 'rpm flag' (flag) is not set.

When the RPM counting completes, it displays the RPM value if it is within the range.

2. Int1 is an interrupt-based function that is called automatically when there is a negative edge on external interrupt 1 pin (P3.3). As soon as the first negative edge arrives, the counter starts counting the external pulses for one second. After one second, the content of TL0 is compared with A6h (166d). For any number greater than 166, multiplication with 60 will yield a number that is longer than four digits, which is out of the display range. This means RPM of max. 9960 (166x60) can be displayed. Otherwise, the LCD will show the message "RPM Out of Range".

3. Display function performs three tasks one by one:

its component layout in Fig. 4.

Operation

1. As the motor starts rotating, the laser light falls on the LDR when the slot aligns with the laser beam and LDR

2. Every time the motor completes one rotation, the monostable (NE555) triggers to generate one pulse, which is indicated by LED1. So LED1 blinks at the rate of motor speed

3. As the first pulse arrives, it generates an interrupt for the microcontroller and immediately the microcontroller starts counting the pulses. This is indicated by LED2. The LCD shows the message "Counting RPM..."

4. The microcontroller counts the pulses for a period of one second. Thereafter, LED2 shows the message